

Solder Card

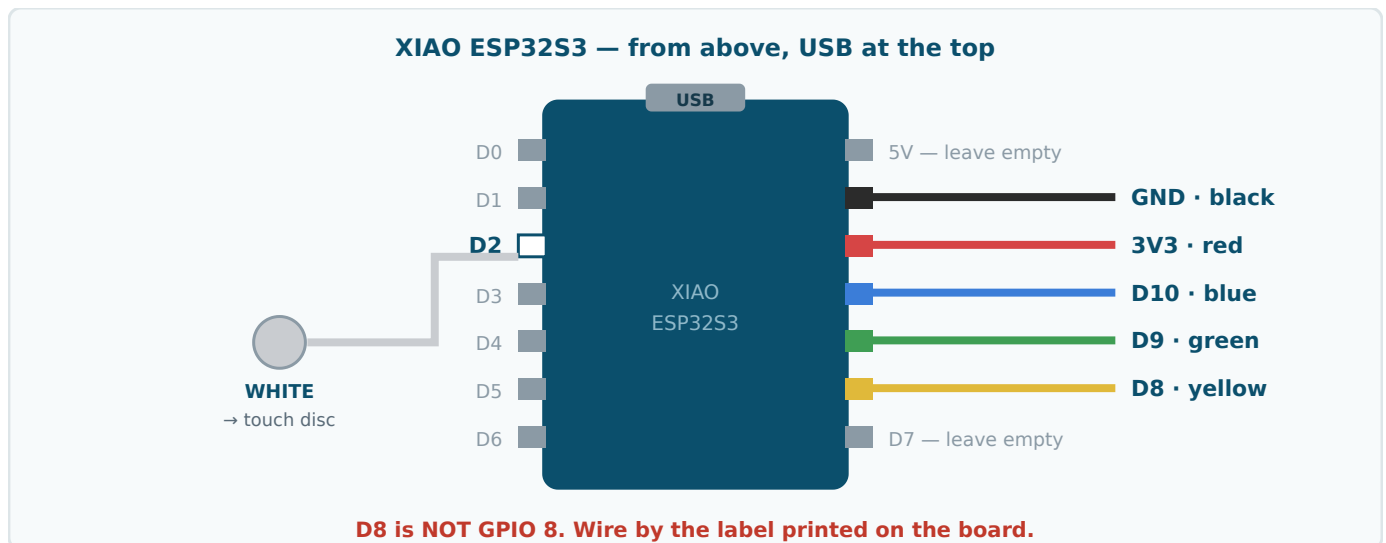
9 joints. Work top to bottom. Tick as you go.

USB UNPLUGGED. Never solder a powered board. Plug in only at page 5.

PREP — BEFORE THE IRON TOUCHES ANYTHING

- ☐ **Cut 5 long wires** — red, black, yellow, green, blue.
Hinge → behind the ear on YOUR glasses, **plus 2cm**. Too short can't be fixed.
- ☐ **Cut 1 white wire, ~5cm**
- ☐ **Strip 3mm** off both ends of all six. Twist each end tight.
- ☐ **Iron to ~350°C**. Tip shiny — if dull or black, wipe it and melt a little solder onto it.
- ☐ **Check your amp module**. Header pins already fitted? Then its 5 connections are **push-on, no iron**.

THE PAD MAP — WIRE BY PAD LABEL, NOT GPIO NUMBER



Five wires in a row on the right edge, one on the left. Nothing else is touched.

ON THE BENCH BEFORE YOU START

- ☐ **Iron + solder** ~350°C
- ☐ **Sponge or brass wool** Wipe the tip between joints
- ☐ **Strippers or a knife** 3mm off each end
- ☐ **Multimeter** **Page 5 needs it** — beep mode

1

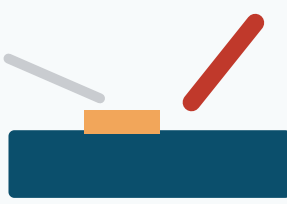
The move — tin, tin, join

Do it in three parts, never in one go

Tape Hold the board down — both hands are busy

A

Tin the PAD

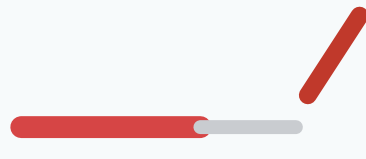


iron on the pad · count 2
then feed solder IN

small shiny dome

B

Tin the WIRE

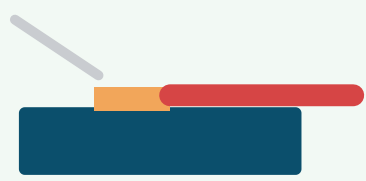


strip 3mm · twist the strands
heat it · solder soaks in

goes stiff and silver

C

Melt them together



hold wire on pad · touch iron
1 second · blobs merge

DON'T MOVE IT FOR 3s

Tinning both sides first turns each joint into a one-second touch instead of a three-handed wrestle.

Heat the PAD, not the solder.

That one sentence is most of learning to solder. Iron on the pad, count two, *then* bring the solder in so the **pad** melts it. Melting solder on the tip and dabbing it on makes a joint that looks fine and falls off later.

READING YOUR OWN JOINTS

What you see	What to do
Shiny, smooth, slightly dished	Good. Move on.
Dull, grey, lumpy	You moved it while cooling. Re-heat till it flows, hold still. Don't add solder.
Ball sat on top, not spread out	Pad was cold. Re-heat the pad properly and it'll wet.
Blob bridging two pads	Fix now. Drag a clean hot tip across it, or use braid.

Work top to bottom down the right edge. That way the iron never has to reach across a joint you've already made.

✓	#	PAD	WIRE	Other end goes to
☐	1	GND	● Black	Amp GND
☐	2	3V3	● Red	Amp VIN
☐	3	D10	● Blue	Amp DIN
☐	4	D9	● Green	Amp LRC / WS
☐	5	D8	● Yellow	Amp BCLK / SCK
☐	6	D2 (left edge)	○ White	Touch disc — joint 9

CHECK FOR BRIDGES — LOOK ALONG THE EDGE, NOT DOWN AT IT

from above — looks fine



three separate pads

you cannot see a bridge from here

from the side — obvious

solder joining two pads



put your eye level with the board

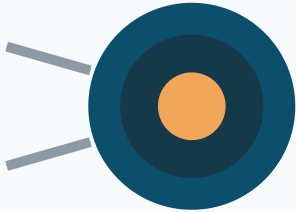
do this after every joint

Tug every wire gently before you move on. A joint that pulls off on the bench will definitely pull off on someone's face.

Keep the blobs flat. The pod only has 10mm of height. If a joint stands proud, re-heat and let it settle rather than forcing the lid down on it.

Found a bridge? Wipe the tip clean, then drag it firmly across the bridge *away* from the pads. The solder follows the hot tip off the board. Re-check from the side. Don't add flux-less solder trying to "push" it apart — that makes it bigger.

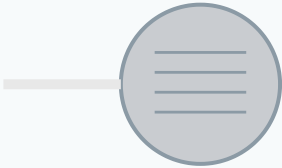
JOINTS 7 & 8



SPEAKER

polarity doesn't matter

JOINT 9

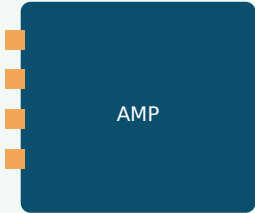


COIN / SCREW HEAD

SCUFF IT WITH SANDPAPER FIRST

solder won't stick to a shiny coin

AMP — PROBABLY NO IRON



header pins fitted?

use push-on jumpers

✓	#	What
☐	7	Short wire → speaker terminal one
☐	8	Short wire → speaker terminal two
☐	9	White wire's free end → the scuffed metal disc
☐	—	Speaker's two free ends → amp OUT+ and OUT-
☐	—	The five bundle wires → amp VIN GND BCLK LRC DIN

Leave **GAIN** , **SD** and **SHUTDOWN** **unconnected**. They have sensible defaults. **SD** usually means *shutdown* — grounding it silences the amp completely, and that is a very confusing hour to lose.

Amp label variants — all the same signals:
BCLK = **SCK** = **CLK** · **LRC** = **WS** = **LRCLK** · **DIN** = **SD** (data) = **DATA**

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Check before you plug in

Two minutes. Do not skip it.

A bridge between 3V3 and GND is a dead short.

It can kill the board the instant you plug in. Find it with a meter, not with smoke.

MULTIMETER ON BEEP MODE — STILL UNPLUGGED

✓	Touch these two	Expected
☐	3V3 and GND	MUST NOT BEEP
☐	D8 and D9	must not beep
☐	D9 and D10	must not beep
☐	D10 and 3V3	must not beep
☐	Each pad → its own far wire end	must beep

Any beep in the first four rows = a solder bridge. Find it, drag a clean hot tip across it, re-test. Don't plug in until all four are silent.

THEN, AND ONLY THEN

- 1 **Plug in USB.**
- 2 **Touch the board and amp within 5 seconds.** More than mildly warm — **unplug immediately**, re-check for bridges.
- 3 Board shows up on your laptop → flash it.
- 4 Camera test first, audio second. Never both at once.

IF SOMETHING'S WRONG

Symptom	Most likely cause
Board won't appear on laptop	Unplug what you soldered. Appears bare → you have a short.
Camera never starts	PSRAM, not wiring — already fixed in the config.
No sound at all	Meter amp VIN → GND, expect ~3.3V. If 0V, joints 1-2.
Static or buzzing only	Yellow and green swapped. D8 →BCLK, D9 →LRC.
Silence but wiring checks out	An SD /shutdown pin pulled low. Disconnect it.
Touch misbehaves	Fires by itself → tune TOUCH_THRESHOLD while worn. Never fires → disc wasn't scuffed.